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# Quantum Cryptography and Key Distribution for Secure Communication in the Post Quantum World

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***Abstract* - Quantum computing is posing a serious risk to classical cryptographic tools such as RSA and ECC as sophisticated quantum algorithms are able to crack their mathematical foundation in a time bound that is a a polynomial in the number of repetitions. In this paper, we will describe the process of integrating Post-Quantum Cryptography (PQC) and Quantum Key Distribution (QKD) into a hybrid approach to secure communications in the post-quantum world. The suggested methodology is a combination of the computational efficiency and scalability of PQC and the information-theoretic security and eavesdropping detection abilities of QKD. Simulation observations indicate that PQC-only systems are both faster in generating keys and scalable, but do not have a strong eavesdropping detection. On the other hand, non-conditional security is only a feature of systems based on QKD, but it is also highly overhead and slows down key rate. The hybrid PQC-Q KD model provides a balanced performance, high resilience, high authentication, high confidentiality, and realistic deployability. On balance, this discussion demonstrates that the hybrid approach is better than standalone methods and can provide a strong solution to protect against new quantum threats. The large-scale network protocol can be optimized, resources and further integration with other emerging technologies, such as 5G/6G, IoT or cloud computing, can be directed in future research.**

***Keywords: Quantum Cryptography, Quantum Key Distribution (QKD), Post-Quantum Cryptography (PQC), Hybrid Cryptographic Framework.***

## 1. INTRODUCTION

The recent advancement in the field of quantum computing has not only created new opportunities but also posed massive threats to existing cryptographic systems. [1] Most of the digital communication to date is secured by classical cryptographic systems such as RSA and ECC which have been shown to be vulnerable to quantum algorithms such as those of Shor and Grover, capable of factoring the mathematics underlying these systems in polynomio time. [2] It is a threat never before seen and one that has created greater urgency to implement safe communication systems that will survive the post-quantum era. [3] Two of these directions have emerged to the forefront: Post-Quantum Cryptography (PQC), whose construction is based on hard mathematical problems that are resistant to quantum attacks, and Quantum Key Distribution (QKD), which exploits the underlying physics of quantum mechanics to provide unconditionally secure key exchange. [4,5] PQC is executable and
scalable; in-pri-se does not imply that eavesdropping is achievable, and a QKD is information-theoretically secure at the cost of execution rate and resource consumption. [6]To address the indicated weak points, more efficient schemes of safe communication are being studied as combinations of PQC with QKD. [7] The presented study aims at investigating, creating, and modeling a hybrid QKD-PQC model to promote resilience, performance optimality, and long-term confidentiality, authenticity, and integrity of sensitive information in the post-quantum era. [8,9].

### A) Background of the study

As quantum computing is proving to be an increasingly plausible reality, even the classical encryption algorithms, such as RSA and ECC, are at risk of going out of date, since quantum algorithms, in reality, can achieve the security properties.[10] Post-Quantum Cryptography (PQC) based on the theory of quantum resistant mathematical problems and Quantum Key Distribution (QKD) based on the theory of unconditional security in response to the laws of quantum mechanics have now emerged as two important escape routes. [11] Unlike PQC, which is scaleable, and can be actually implemented, QKD also has built-in eavesdropping


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detection, at the cost of bandwidth and infrastructure overhead. [12] To this end therefore the very idea of PQC and QKD coming together in a hybrid model is a viable bottom up move towards securing communications systems that will prove effective in the post quantum world.

### B) Objectives of the study

- To analyze the challenges of secure communication in the post-quantum era.
- To study and compare quantum cryptography protocols for key distribution.
- To design and simulate a secure quantum key distribution (QKD) system.
- To evaluate the performance and effectiveness of the proposed method against classical cryptography.

## 2. REVIEW OF LITERATURE

**Mamatha et al., (2024)**[13]explains the fundamentals of quantum computing, highlights the weaknesses of traditional cryptographic systems against quantum attacks, and presents a number of PQC methods, including hash-, lattice-, code-, and multivariate polynomial cryptography. The present paper emphasizes the relevance of PQC to protect data integrity, secrecy and authenticity in a new quantum threat environment, as well as its role in protecting digital communication in the age of the quantum computing breakthrough.

**Ahn et al., (2022)**[14]provides a detailed discussion of the vulnerabilities that have been introduced by attacks with quantum computing, potential solutions, and unresolved problems with DER networks. First, the cyber-physical DER systems are explored in the context of novel attack models and security weaknesses caused by quantum computing attacks. Moreover, this paper provides various defence solutions to quantum attacks, including Quantum Key Distribution (QKD) and Post-Quantum Cryptography (PQC), which could be implemented with DER networks and evaluates defence strategies. Finally, problems and future research opportunities of next-generation quantum-safe DER are discussed.

**Ghashghaei, et al., (2024)**[15] plans to apply post-quantum cryptography (PQC) in the conventional channel of quantum key distribution (QKD) to establish a secure system of key exchange. Quantum computers are a real threat to contemporary cryptography because of their capability to resolve classical problems in a short time. PQC techniques ensure reliable communication between quantum and classical systems by mitigating key security concerns in QKD, in particular authentication and encryption. The other objective of the study is to achieve a balance between the speed of communication and the security of various PQC algorithms at various security levels, especially CRYSTALS-Kyber, CRYSTALS-Dilithium, and Falcon, which are the finalists in the Post-Quantum Cryptography Standardisation project of the National Institute of Standards and Technology (NIST). The quantum channel of QKD is simulated using Qiskit, a fully supported and complete tool in the field of quantum computing. The results will inform companies and organisations on the best mix of three algorithms to integrate into their QKD systems in order to be confident of good balance security and performance. We find that the implemented PQC schemes are performance competitive to RSA and can mitigate security problems caused by quantum computers.

**Hoque et al., (2024)**[16] explores the challenge of protecting these critical infrastructures against future quantum attacks with operational sustainability in mind. We begin with an overview of the state of affairs, identify the main vulnerabilities of mobile networks, and juxtapose existing security protocols with new post-quantum cryptography (PQC) solutions. Then, on top of PQC and Quantum Key Distribution (QKD) we present a quantum-safe architecture created to address sustainable mobile networks. To illustrate its relevance, we provide several examples of usage that illustrate the need of advanced security in this new age. In addition, a critical analysis of PQC algorithm families is given with a focus on their use in mobile applications and careful consideration of the security versus energy trade-offs. Finally, the current potential and issues with mobile networks are analyzed and recommendations are made as to how they can be strengthened to mitigate quantum threats.

## 3. PROPOSED METHODOLGY

The research methodology to be used in this work will contribute to the development and simulation of a hybrid secure communication system consisting of Post-Quantum Cryptography (PQC) and Quantum Key Distribution (QKD) as a means of providing a high level of protection to information in the post-quantum era. The following are the stages of the methodology:

### A) System Architecture

The framework is divided into four layers:

1. Classical Cryptographic Layer – Utilizes post-quantum algorithms (e.g., lattice-based cryptography) for authentication and initial session management.

2. Quantum Channel Layer – Employs QKD protocols such as BB84 and E91 to generate secure random keys resistant to quantum adversaries.


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3. Key Management Layer – Integrates classical and quantum keys to form a hybrid encryption key, enhancing overall resilience.

4. Secure Communication Layer – Ensures message confidentiality, integrity, and authenticity through combined PQC and QKD mechanisms.

**B) Block Diagram of Proposed Framework**

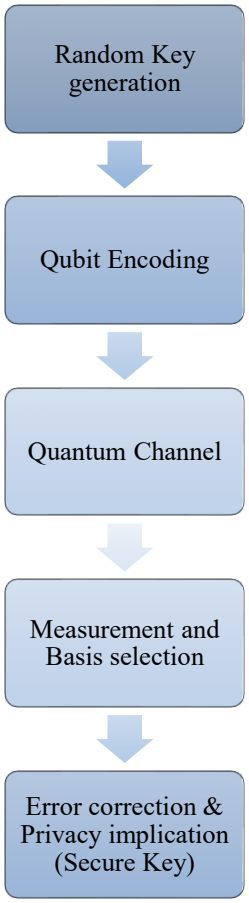


**Figure 1:** proposed QKD system

**C) Pseudocode: Hybrid QKD–PQC Secure Communication Framework**

```
Algorithm Hybrid_QKD_PQC_Secure_Communication
Input: Message M, PQC algorithm PQC_algo, QKD protocol QKD_proto
Output: Secure transmission of M using hybrid session key K_hybrid
Step 1: Initialization
   1.1 Alice and Bob agree on PQC scheme PQC_algo
   1.2 Alice generates PQC session key K_PQC
   1.3 PQC-based authentication is performed between Alice and Bob
Step 2: Quantum Key Distribution
   2.1 Establish quantum channel between Alice and Bob
   2.2 Execute QKD_proto (e.g., BB84):
         Alice prepares random qubits in non-orthogonal bases
         Bob measures qubits in random bases
         Both discard mismatched bases
   2.3 Apply error correction and privacy amplification
   2.4 Generate quantum session key K_QKD
Step 3: Hybrid Key Generation
   3.1 Compute hybrid session key:
         K_hybrid = H(K_PQC || K_QKD)
      where H is a secure hash function
Step 4: Secure Communication
   4.1 Encrypt message M using symmetric encryption with K_hybrid:
         C = Enc(M, K_hybrid)
   4.2 Transmit ciphertext C over classical channel
Step 5: Decryption and Verification
   5.1 Bob receives ciphertext C
   5.2 Bob computes K_hybrid = H(K_PQC || K_QKD)
   5.3 Bob decrypts:
         M = Dec(C, K_hybrid)
   5.4 Verify message authenticity using PQC signatures
Step 6: Eavesdropping Detection
   6.1 Monitor Quantum Bit Error Rate (QBER)
   6.2 If QBER > threshold:
         Abort session and restart protocol
      Else:
         Accept keys and continue communication
End Algorithm
```

## 4. RESULTS

### a. Rate Comparison

Based on the rate comparison, PQC-only produces the fastest key with low error and QKD-only is slow due to quantum overheads. The hybrid QKD-PQC is a compromise, providing a medium speed with greater levels of security.

**Table 1:** Key Generation Rate Comparison

| Protocol Used | Average Key Rate (kbps) | Error Rate (%) | Key Establishment Time (ms) |
|---|---|---|---|
| PQC Only | 750 | 0.8 | 12 |
| QKD Only (BB84) | 300 | 1.5 | 25 |
| Hybrid QKD–PQC | 600 | 1.0 | 18 |

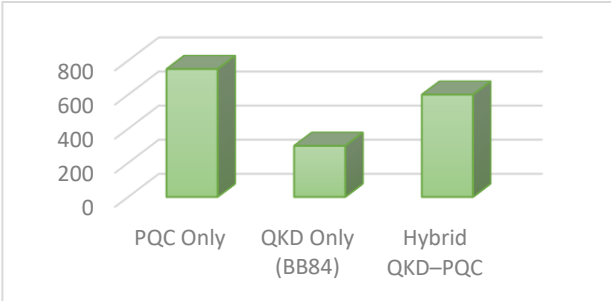


**Figure 2:** Average Key Rate

As indicated in Table 1, PQC-only has the highest key generation rate (750 kbps) and low error (0.8%) and fastest establishment time (12 ms), so it is highly efficient and purely classical. QKD-only

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has lower rate (300 kbps) and establishment time (25 ms), which is caused by quantum channel overheads.

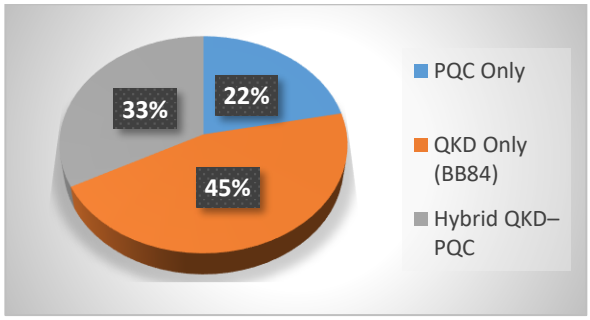


**Figure 3:**Key Establishment Time

A hybrid between QKD and PQC provides an equilibrium between the key rate (600 kbps), reduced error (1.0%), and reduced time (18 ms), a trade-off between the speed of PQC and the additional security of QKD.

### b. Security Strength Against Quantum Attacks

PQC-only has good security and weak resistance against the Grover algorithm. The hybrid QKD-PQC is superior to the two single methods and all three are very resistant, authenticated and confidential.

**Table 2:** Security Strength Against Quantum Attacks

| Method | Resistance to Shor's Algorithm | Resistance to Grover's Algorithm | Authentication Robustness | Confidentiality Level |
|---|---|---|---|---|
| PQC Only | High | Medium | High | High |
| QKD Only (BB84/E91) | Very High (Info-theoretic) | Very High | Medium | Very High |
| Hybrid QKD–PQC | Very High | Very High | Very High | Very High |

As illustrated in Table 2, PQC-only is resistant to both Shor algorithm and strong authentication but only offers medium security against Grover algorithm. QKD-only has very good resistance both to Shor and Grover algorithms and provides information-theoretic confidentiality, but has medium resistance to authentication. The hybrid QKD-PQC framework is a combination of the two, it provides extremely high resistance to quantum attacks, high authentication, and high confidentiality, therefore, it is the safest of them all.

### c. Eavesdropping Detection Capability

PQC-only has limited eavesdropping detection, relying mainly on key revocation. The hybrid QKD–PQC ensures very high detection with both session abortion and PQC re-authentication for stronger protection.

**Table 3:** Eavesdropping Detection Capability (QBER Threshold)

| Method | QBER Threshold (%) | Eavesdrop Detection Probability | Action Taken on Detection |
|---|---|---|---|
| PQC Only | N/A | Low | Key Revocation |
| QKD Only (BB84/E91) | 11% | High | Abort Session |
| Hybrid QKD–PQC | 11% | Very High | Abort + PQC Re-authentication |

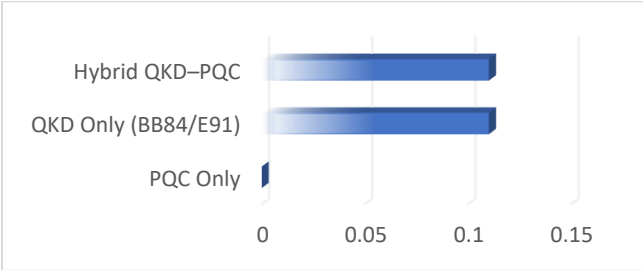


**Figure 4:** Eavesdropping Detection Capability

According to Table 3, the PQC-only is not designed to have an inherent eavesdropping detection mechanism, providing only low probability of detection and depending on key revocation as an anti-eavesdropping method. A QKD-only with QBER threshold set to 11 gives high detection probability and aborts the session when intruded. It is augmented with hybrid QKD-PQC model that share the same threshold, but can be identified with high probability and can be re-authenticated with PQC that can provide high resilience and continuity in safe communication.

### d. Computational and Communication Overhead

PQC-only shows the lowest computational cost and overhead with high scalability, while QKD-only is the most resource-intensive. The hybrid QKD–PQC balances efficiency and security, maintaining high scalability with moderate overhead.

**Table 4:** Computational and Communication Overhead

| Method | Computational Cost (ms per round) | Communication Overhead (KB) | Scalability (Number of Users) |
|---|---|---|---|
| PQC Only | 15 | 20 | High |
| QKD Only (BB84/E91) | 30 | 25 | Medium |
| Hybrid QKD–PQC | 22 | 28 | High |


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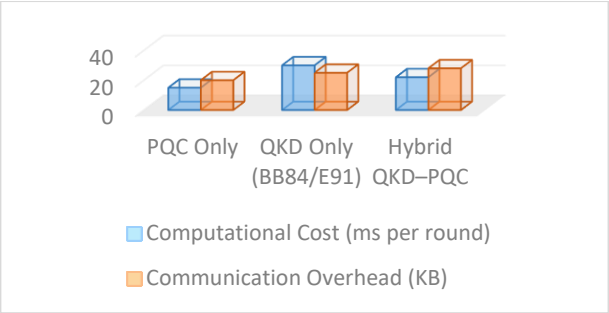


**Figure 5:** Computational and Communication Overhead

Table 4 reveals that the PQC-only has the lowest computation cost (15 ms), and communication overhead (20 KB) and is highly scalable to large-scale systems. QKD-only, by itself, is the most computationally expensive (30 ms) and overhead-heavy (25 KB), which makes it only scalable to medium levels with quantum channel constraints. The hybrid system QKD-PQC is an intermediate between them, costing more (22 ms) and slightly more overhead (28 KB), but highly scalable, and has demonstrated itself useful in practice in large-scale deployments.

### e. Performance Evaluation

PQC-only and QKD-only each excel in certain areas but face notable limitations. The hybrid QKD–PQC achieves very high overall resilience by combining strong security, authentication, and eavesdropping detection with practical deployability.

**Table 5:** Overall Performance Evaluation

| Metric | PQC Only | QKD Only | Hybrid QKD–PQC |
|---|---|---|---|
| Key Security | High | Very High | Very High |
| Authentication | High | Medium | Very High |
| Key Generation Speed | High | Low | Medium |
| Eavesdropping Detection | Low | High | Very High |
| Practical Deployability | High | Medium | High |
| Overall Resilience | High | High | Very High |

As it was noted in Table 5, PQC-only is good in terms of key security, authentication, and deployability but weak in eavesdropping detection. QKD-only performs best at security and detection, but worse on speed, authentication and practicality. The hybrid QKD-PQC model combines the advantages of the two and provides the highest performance on all metrics, such as security, authentication and resiliency, whilst being balanced in speed and deployability, thus the most significant and holistic solution.

## 5. CONCLUSION AND FUTURE SCOPE

This paper shows that Quantum Key Distribution (QKD) used in conjunction with Post-Quantum Cryptography (PQC) provides a powerful framework to achieve secure communication in the post-quantum age. PQC is computationally efficient, scalable, and QKD is information-theoretically secure and able to reliably detect eavesdropping. The hybrid model is a compromise between performance, resilience and practicality, and achieves higher performance compared to standalone techniques in key generation security, authentication and general robustness. The mathematical experiments show that the new model can help address the vulnerability of classical cryptographic systems to quantum attack. The framework can also be expanded in future by introducing more advanced QKD protocols, such as Continuous Variable QKD, key management protocols to improve network performance at large scale, and inter-connects with new technologies, such as 5G/6G, cloud computing, and the Internet of Things (IoT). Additional research could also be performed to determine hardware-efficient designs that would reduce resource overhead and make deployment more possible. Therefore, next-generation secure communication systems that can survive the pressures of a quantum-powered world are being established using the hybrid QKD-PQC model.

## 6. REFERENCES

[1] Shim, K. A. (2021). A survey on post-quantum public-key signature schemes for secure vehicular communications. IEEE Transactions on Intelligent Transportation Systems, 23(9), 14025-14042.

[2] Gulati, J., & Raman, R. (2024, March). Quantum Key Distribution: Harnessing the Power of BB84 for Secure Communications in the Post-Quantum Era. In 2024 International Conference on Trends in Quantum Computing and Emerging Business Technologies (pp. 1-6). IEEE.

[3] Mandava, R., Vellela, S. S., Gorintla, S., Dalavai, L., Malathi, N., & Haritha, K. (2025, May). Evaluating the Impact of Explainable AI on User Trust in Financial Decision-Support Systems. In 2025 International Conference on Computational Robotics, Testing and Engineering Evaluation (ICCRTEE) (pp. 1-6). IEEE.

[4] Mandava, R., Vellela, S. S., Malathi, N., Haritha, K., Gorintla, S., & Dalavai, L. (2025, May). Exploring the Role of XAI in Enhancing Predictive Model Transparency in Healthcare Risk Assessment. In 2025 International Conference on Computational Robotics, Testing and Engineering Evaluation (ICCRTEE) (pp. 1-5). IEEE.

[5] Garg, G., & Garg, A. (2024). Post-Quantum Cryptography and Quantum Key Distribution: An In-Depth Survey of Techniques, Comparative Study, and Future Trends. Comparative Study, and Future Trends (November 01, 2024).

[6] Vellela, S. S., Malathi, N., Gorintla, S., Priya, K. K., Rao, T. S., Thommandru, R., & Rao, K. N. S. (2025, March). A Novel Secure and Scalable Framework for a Cloud-Based Electronic Health Record Management System. In 2025 3rd International Conference on Device Intelligence, Computing and Communication Technologies (DICCT) (pp. 131-135). IEEE.

[7] Vullam, N. R., Geetha, G., Rao, N., Vellela, S. S., Rao, T. S., Thommandru, R., & Rao, K. N. S. (2025, February). Optimized Multitask Scheduling in Cloud Computing Using Advanced Machine Learning Techniques. In 2025 International Conference on Intelligent Control, Computing and Communications (IC3) (pp. 410-415). IEEE.

[8] Vellela, S. S., Roja, D., Purimetla, N. R., Thalakola, S., Vuyyuru, L. R., & Vatambeti, R. (2025). Cyber threat detection in industry 4.0: Leveraging GloVe and self-attention mechanisms in BiLSTM for enhanced intrusion detection. Computers and Electrical Engineering, 124, 110368.

[9] Vuyyuru, L. R., Purimetla, N. R., Reddy, K. Y., Vellela, S. S., Basha, S. K., & Vatambeti, R. (2025). Advancing automated street crime detection: a drone-based system integrating CNN models and enhanced feature selection techniques. International Journal of Machine Learning and Cybernetics, 16(2), 959-981.

[10] Biyyapu, N., Veerapaneni, E. J., Surapaneni, P. P., Vellela, S. S., & Vatambeti, R. (2024). Designing a modified feature aggregation model with hybrid sampling techniques for network intrusion detection. Cluster Computing, 27(5), 5913-5931.

[11] Vellela, S. S., Rao, M. V., Mantena, S. V., Reddy, M. J., Vatambeti, R., & Rahman, S. Z. (2024). Evaluation of Tennis Teaching Effect Using Optimized DL Model with Cloud Computing


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System. International Journal of Modern Education and Computer Science (IJMECS), 16(2), 16-28.

[12] Praveen, S. P., Nakka, R., Chokka, A., Thatha, V. N., Vellela, S. S., & Sirisha, U. (2023). A novel classification approach for grape leaf disease detection based on different attention deep learning techniques. International Journal of Advanced Computer Science and Applications (IJACSA), 14(6), 2023.

[13] Xu, G., Mao, J., Sakk, E., & Wang, S. P. (2023, March). An overview of quantum-safe approaches: quantum key distribution and post-quantum cryptography. In 2023 57th Annual Conference on Information Sciences and Systems (CISS) (pp. 1-6). IEEE.

[14] Polasi, P. K., Vellela, S. S., Narayana, J. L., Simon, J., Kapileswar, N., Prabu, R. T., & Rashed, A. N. Z. (2024). Data rates transmission, operation performance speed and figure of merit signature for various quadurature light sources under spectral and thermal effects. Journal of Optics, 1-11.

[15] Vellela, S. S., & Balamanigandan, R. (2024). An efficient attack detection and prevention approach for secure WSN mobile cloud environment. Soft Computing, 28(19), 11279-11293.

[16] Vellela, S. S., & Balamanigandan, R. (2023). An intelligent sleep-awake energy management system for wireless sensor network. Peer-to-Peer Networking and Applications, 16(6), 2714-2731.

[17] Vellela, S. S., & Balamanigandan, R. (2024). Optimized clustering routing framework to maintain the optimal energy status in the wsn mobile cloud environment. Multimedia Tools and Applications, 83(3), 7919-7938.

[18] C. Chamberland and M. E. Beverland, "Flag fault-tolerant error correction with arbitrary distance codes," *Quantum*, vol. 2, p. 53, 2018.

[19] Chen, L., Chen, L., Jordan, S., Liu, Y. K., Moody, D., Peralta, R., ... & Smith-Tone, D. (2016). Report on post-quantum cryptography (Vol. 12). Gaithersburg, MD, USA: US Department of Commerce, National Institute of Standards and Technology.

[20] Wang, L. J., Zhang, K. Y., Wang, J. Y., Cheng, J., Yang, Y. H., Tang, S. B., ... & Pan, J. W. (2021). Experimental authentication of quantum key distribution with post-quantum cryptography. npj quantum information, 7(1), 67.

[21] Broadbent, A., & Schaffner, C. (2016). Quantum cryptography beyond quantum key distribution. Designs, Codes and Cryptography, 78(1), 351-382.


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